1. 🏠 Project Documentation: Visualizing Housing Market Trends

Project Objective: To address challenges in understanding factors influencing house prices and sales trends through data-driven insights to optimize pricing strategies and market competitiveness.

---

2. PROJECT DESCRIPTION

- **System Overview:** An end-to-end analytical ecosystem and web integration platform designed to convert raw housing datasets into an interactive, high-fidelity visual storyboard.
- **Problem Domain:** Real Estate Business Intelligence (BI) and Market Trend Forecasting.
- **Core Technology Stack:** Python (Flask), Tableau, and Modern Frontend Technologies (HTML5, CSS3, JavaScript).
- **Key Capabilities:** High-performance interactive dashboarding, automated data pipeline cleaning, stress-tested multi-parameter filtering, and narrative-driven data storytelling.
- **Target Users:** Strategic real estate analysts, data-driven marketing teams, and C-suite executives at ABC Company.
- **Real-world Relevance:** Provides a competitive edge by isolating property value drivers—such as renovation timelines and structural configurations—to refine pricing models and market positioning.

---

3. APPLICATION SCENARIOS / USE CASES

- **Scenario 1: Overall Data Overview:** Delivers a comprehensive snapshot of the dataset's scale, featuring ~21,000 unique records, a calculated average sale price of **$511,619**, and total basement area metrics.
- **Scenario 2: Total Sales by Years Since Renovation:** Utilizes a frequency distribution (histogram) to evaluate how renovation age impacts sales volume across specific price brackets, notably identifying a market peak in the **$450k range**.
- **Scenario 3: Distribution of House Age by Renovation Status:** Employs a categorical pie chart to assess inventory age demographics and the historical prevalence of renovations across property generations.
- **Scenario 4: Structural Feature Correlation:** A grouped bar chart analysis correlating property age with critical physical attributes: number of bathrooms, bedrooms, and floor levels.

---

4. TECHNICAL ARCHITECTURE OVERVIEW

- **Frontend Layer:** Developed using HTML, CSS, and JS to provide a responsive host environment for the embedded Tableau Storyboard.
- **Backend Layer:** Orchestrated by **Flask (Python)** to manage server-side routing and the seamless delivery of integrated components.
- **External Services:** Leverages **Tableau Public/Desktop** for advanced data visualization processing and real-time interactive filtering.
- **Deployment Environment:** Currently live via **GitHub Pages** for frontend delivery, with a roadmap for full-stack cloud migration.

---

5. PREREQUISITES

5.1 Software Requirements

- **Integrated Development Environment (IDE):** Visual Studio Code (VS Code) or PyCharm.
- **Runtime Environment:** Python 3.x.
- **Visualization Platform:** Tableau Desktop or Tableau Public Edition.

5.2 Libraries / Frameworks / Dependencies

- **Flask:** Primary web micro-framework for system integration.
- **Pandas & NumPy:** Essential for data manipulation and preprocessing.

5.3 Hardware Requirements

- **System Specifications:** Minimum 8GB RAM and an Intel i5 processor (or equivalent) to facilitate smooth rendering of high-volume Tableau datasets.

---

6. PRIOR KNOWLEDGE REQUIRED

- **Programming:** Proficiency in Python syntax and Flask-based routing.
- **Data Science:** Fundamentals of data cleaning, outlier detection, and normalization.
- **Data Visualization:** Mastery of Tableau dimensions, measures, calculated fields, and dashboard actions.

---

7. PROJECT OBJECTIVES

- **Technical:** Achieve seamless embedding of multi-scenario Tableau dashboards into a stable Flask web interface.

- **Performance:** Optimize the system to maintain responsive filtering speeds across a 21,000+ record set.
- **Deployment:** Sustain 100% availability for the live stakeholder demonstration.
- **Learning Outcomes:** Successfully implement the full Agile project lifecycle, spanning from raw data extraction to public web publishing.

---

8. SYSTEM WORKFLOW

1. **User Access:** Stakeholders navigate to the live URL hosted on GitHub Pages/Flask.
2. **Input Capturing:** Users apply dynamic filters (e.g., House Age, Renovation Status) via the UI.
3. **Visualization Processing:** The Tableau engine computes the filtered logic in real-time.
4. **Backend Management:** Flask serves the optimized views while maintaining UI responsiveness.
5. **Dynamic Rendering:** The dashboard updates instantly to reflect specific sales bins and demographic distributions.
6. **Insights Delivery:** Users interpret visual data to inform strategic real estate decisions.

---

9. MILESTONE 1: REQUIREMENT ANALYSIS & SYSTEM DESIGN

- **Problem Definition:** ABC Company identified a lack of granular visibility into how specific renovation and structural variables drive ROI.
- **Functional Requirements:** Automated data extraction, comprehensive cleaning, 4-scenario visualization, and secure web integration.
- **Non-Functional Requirements:** Prioritized rendering speed, mobile-friendly design, and data integrity.
- **Tech Justification:** Tableau was selected for its superior capability in handling complex statistical distributions (histograms and pie charts) compared to static plotting libraries.

---

10. MILESTONE 2: ENVIRONMENT SETUP & INITIAL CONFIGURATION

- **Environment:** Verified installation of the Python interpreter, Flask environment, and Tableau Public.
- **Standardized Repository Structure:**
- Visualizing-Housing-Market-Trends/
- │StoryBoard
- ├── Web Integration/
- │ ├── index.html # Primary Frontend UI

- │ └── temp.py # Flask Backend Script
- ├── Data/
- │ └── Transformed_Housing_Data.csv # Cleaned Dataset
- └── README.md # Project Documentation

---

11. MILESTONE 3: CORE SYSTEM DEVELOPMENT

- **Feature 1 (Data Acquisition): Arya Batham** executed the collection and extraction of the primary housing dataset from the source database.
- **Feature 2 (Data Refinement): Khushi Tomar** performed advanced outlier handling and formatted the data schema for Tableau compatibility.
- **Feature 3 (Visual Analytics):** Developed the "Comprehensive House Data Analysis" dashboard, focusing on sales performance and property demographics.

---

12. MILESTONE 4: INTEGRATION & OPTIMIZATION

- **System Integration: Ojasv Singh** engineered the embedding of Tableau Stories into the Flask-driven web environment.
- **Performance Optimization: Nishant Vishwakarma** tuned the filter utilization and reduced data load latency to ensure a fluid user experience.

---

13. MILESTONE 5: TESTING & VALIDATION

Test Case ID	**Input**	**Expected Output**	**Status**
TC01	Apply "Renovated" Filter	Charts dynamically update to isolate renovated properties	**Passed**
TC02	Initial 21k Record Load	Dashboard becomes fully interactive in <3 seconds	**Passed**

---

14. DEPLOYMENT

- **Hosting Strategy:** Leveraging **GitHub Pages** for the UI/frontend and local/cloud instances for the Flask logic.
- **Execution Steps:**
    1. Push codebase to the GitHub repository.
    2. Configure GitHub Pages within the repository settings.
    3. Define the entry point as `Web Integration/index.html`.

---

15. RESULTS

- **Key Findings:** The analysis definitively identified the **$450k price point** as the peak sales bin and confirmed **bedroom count** as the dominant value driver for properties in the 50-year age bracket.
- **Strategic Evaluation:** The system provides a robust platform for high-volume data storytelling, turning raw metrics into actionable business intelligence.

---

16. ADVANTAGES & LIMITATIONS

- **Advantages:** Exceptional user interactivity, narrative-focused Storyboarding, and zero-cost web accessibility for stakeholders.
- **Limitations:** Real-time visual availability is currently coupled with Tableau Public server uptimes.

---

17. FUTURE ENHANCEMENTS

- **Automation Pipeline:** Implementing live web scraping to facilitate real-time data refreshes.
- **Cloud Scaling:** Full enterprise deployment on **AWS** or **Heroku**.
- **Predictive Modeling:** Integrating **Scikit-Learn** for ML-driven future price forecasting based on current trends.

---

18. CONCLUSION

The project successfully delivered a specialized analytical tool for ABC Company. By synthesizing Tableau's visualization depth with the flexibility of the Flask framework, the team created a high-impact platform that empowers stakeholders to optimize pricing and market strategies through empirical property data.

---

19. APPENDIX

- **GitHub Repository:** Visualizing-Housing-Market-Trends
- **Live Demo:** Interactive Market Dashboard
- **Video Presentation:** Video Link